

Changyen (Brian) Li

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Education

University of Texas at Austin

Aug. 2025 – May 2027

M.S. Electrical and Computer Engineering (Software Engineering and Systems Track)

- GPA: **3.95/4.0**, Courses: Advanced Database Systems, Parallel Programming, Machine Learning Hardware

National Taiwan University (NTU)

Jun. 2024

B.S. Computer Science and Information Engineering

- GPA: **3.99/4.0**, Class Rank: **2/145**, 3 **Dean's List Awards**
- Relevant Courses: Operating Systems (OS), Computer Networks, Algorithm Design and Analysis, Machine Learning, Full Stack Web Development, Unix Systems Programming, Cryptography, Computer Architecture

Work Experience

Software Engineer Intern

Jun. 2026 – Aug. 2026

Google

- Architected and launched **Penja Snapshot**, a one-click workflow that captures and replays complete device states for issue triage on Google Home's debugging platform used by **400+ developers** weekly
- Saved an estimated **30+ developer-hours weekly** across debugging workflows for **20+** current and legacy Google products by shipping a full-stack snapshot system spanning versioned **Protocol Buffers** contract, **Java/gRPC** backend, and **TypeScript/Angular** frontend for point-in-time and time-series recording
- Reduced LLM context usage by **70%** and agent steps by **78%** versus screenshot inputs in benchmarks using real production bug tickets by structuring device state as agent-searchable JSON for progressive retrieval

Software Engineer Intern

Jan. 2023 – Jun. 2023

Synology Inc.

- Eliminated 3 hours of manual effort per company-wide registry audit across 10,000+ self-hosted GitLab repositories by building a fault-tolerant **Python** and **GitLab CI/CD** pipeline with bounded concurrency and automatic retries
- Cut self-hosted GitLab registry storage by 20%+ within two weeks of launch by automating digest-aware image cleanup with configurable retention policies and staged owner notifications, while delivering registry usage telemetry via **OpenSearch** to guide long-term infrastructure capacity planning

Projects

Nanologger

- Built a lock-free SPSC asynchronous binary logger in C++20. The producer path never allocates, locks, formats or makes syscalls, and formatting and I/O run on a pinned consumer thread. It reaches 18 ns P50, 32 ns P99 per log and cuts equal-work P99 latency by 5.6x compared with `spdlog`.
- Cut producer P99 latency by 61%, L1 cache misses by 89%, and run-to-run P50 jitter from 32 ns to <1 ns by using `perf` to trace false sharing on the ring cursors and moving each thread's private state onto its own cache line.

Cloud-Native Order Processing Platform

- Architected and deployed a cloud-native order-processing platform on Amazon EKS and RDS using Go, GraphQL, gRPC, PostgreSQL, and Kubernetes, sustaining 200+ orders/sec at sub-20 ms p95 latency across 100,000+ requests with zero errors or correctness violations.
- Engineered idempotent payment workflows, atomic inventory reservation, compensating transactions, and Kubernetes autoscaling, preventing overselling and duplicate charges under concurrent load while scaling from 1 to 4 replicas during a 5x traffic spike with zero downtime through rolling deployments.

Awards and Honors

- **ICPC Asia Regional Contests**: Earned **2 Gold** and **1 Silver** Awards
- **Taiwanese Olympiad in Informatics**: Ranked **24th nationally** in Taiwan's IOI team selection
- Solved complex math, algorithm, and data structure problems under strict time limits using C++ and Python

Skills

- **Programming Languages**: C, C++, Python, Java, JavaScript, TypeScript
- **Web Development**: React.js, Next.js, Angular, OAuth, PostgreSQL, LDAP
- **Cloud & DevOps**: AWS, Docker, Kubernetes, CI/CD, Git, Linux